

Evaluation Report

For the performance metrics, Buyer and Seller clients were simulated by spawning them on different threads. For every run, the Buyer and the Seller thread first run the register account operation and then the login operation. After that, for the rest of the runs, the buyer client iterates through the following actions: "PRODUCT_SEARCH", "GET_BUYER_PURCHASES", "SAVE_CART", "CLEAR_CART", "DISPLAY_CART", and the seller client iterates through the following actions: "GET_RATING", "REGISTER_PRODUCT", "CHANGE_ITEM_PRICE", and "GET_RATING". At the end of each run, the clients perform the logout operation.

The operations were performed on the local machine: MacBook Pro - 2020 model (i5 10th Gen, quad core with 16 GB RAM.)

This machine proved to be quite low-powered for the operations with 10 concurrent clients and onwards. All these scenarios involved 10 runs and 1000 calls each.

Scenario	Average Response Time (ms)	Throughput
1 Concurrent Buyer and Seller	6.31	279 calls/sec
10 Concurrent Buyer and Seller	32.69	525 calls/sec
100 Concurrent Buyer and Seller	56.89	3460 calls/sec

As we move from 1 concurrent Buyer and Seller to 10, we can see that the average response time increases as the resources are managed between 20 threads. This also results in increased throughput.

As we moved from scenario 1 to 2 to 3, the number of timeout errors increased.